Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	**TE-AI&DS**	**Semester:**	**V**
Course Code:	**2015111**	**Course Name:**	**Statistics for Machine Learning and Data Science Lab**

Name of Student:	**ASHLESHA .S. BHOIR**
Roll No. :	**03**
Experiment No.:	**2**
Title of the Experiment:	**Apply Descriptive Statistics and Data Visualization Techniques for Exploratory Data Analysis**
Date of Performance:	
Date of Submission:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty : **Dr. Tatwadarshi Nagarhalli**

Signature :

Date :

Aim: To apply descriptive statistical measures and data visualization techniques to summarize, analyze, and interpret the distribution and relationships present in the Pima Indians Diabetes Dataset.

Objectives

After completing this experiment, the student will be able to:

- Apply descriptive statistical and visualization techniques to explore the characteristics of a real-world healthcare dataset.
- Interpret distributions, variability, relationships, and patterns obtained from statistical analysis and visualizations.

Tools Required

1. Python 3.x
2. Google Colab / Jupyter Notebook
3. Pandas
4. NumPy
5. Matplotlib
6. Seaborn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for female patients and is used to study factors associated with diabetes.

The dataset contains 768 observations and 8 input attributes, along with a binary outcome variable indicating diabetes status.

Attribute	Description

Pregnancies	Number of pregnancies
Glucose	Plasma glucose concentration
BloodPressure	Diastolic blood pressure
SkinThickness	Triceps skin fold thickness
Insulin	2-Hour serum insulin
BMI	Body Mass Index
DiabetesPedigreeFunction	Diabetes pedigree function
Age	Age in years
Outcome	1 = Diabetic, 0 = Non-diabetic

Procedure

1. Load the Pima Indians Diabetes Dataset into a Pandas DataFrame.
2. Inspect the dataset structure, dimensions, column names, and data types.
3. Calculate descriptive statistics including mean, median, mode, minimum, maximum, variance, and standard deviation for selected numerical variables.
4. Analyze the distribution of variables using histograms and boxplots.
5. Analyze the frequency of the Outcome variable using a bar chart.
6. Generate scatter plots to study relationships such as Glucose vs. BMI and Age vs. Glucose.
7. Generate a pair plot for selected numerical variables.
8. Interpret the statistical summaries and visualizations and record important observations.

Description of the Experiment

Descriptive statistics summarize the important characteristics of a dataset using numerical measures, while data visualization presents these characteristics graphically.

The experiment follows the workflow: Dataset, Descriptive Statistics, Distribution Analysis, Visualization, Relationship Analysis, Interpretation,

Students will use the Pima Diabetes Dataset to analyze the central tendency, variability, distributions, and relationships among health-related variables.

Detailed Description of the Statistical techniques:

Mean, Median, and Mode

These measures describe the central or typical value of a dataset.

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

The median is the middle value, while the mode is the most frequently occurring value.

Range, Variance, and Standard Deviation

These measures describe data variability.

$$Range = Maximum - Minimums^2 = \frac{\sum(x_i - \bar{x})^2}{n-1} s = \sqrt{s^2}$$

Histogram

Used to study the distribution, spread, and shape of numerical variables such as Glucose, BMI, and Age.

Boxplot

Used to visualize the median, quartiles, spread, and potential outliers.

$$IQR = Q_3 - Q_1$$

Bar Chart

Used to display the frequency of categorical or binary variables such as Outcome.

Scatter Plot

Used to examine the relationship between two numerical variables, such as:

- Glucose and BMI
- Age and Glucose

Pair Plot

Used to simultaneously visualize pairwise relationships and distributions among multiple numerical variables.

Implementation

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from google.colab import files

uploaded = files.upload()
```

```
Choose Files diabetes.csv
diabetes.csv(text/csv) - 23873 bytes, last modified: 7/29/2026 - 100% done
Saving diabetes.csv to diabetes.csv
```

```
import os

print(os.listdir())
```

```
['.config', 'diabetes.csv', 'sample_data']
```

```
import pandas as pd

df = pd.read_csv("diabetes.csv")
```

```
df.head()
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunct
0	6	148	72	35	0	33.6	0
1	1	85	66	29	0	26.6	0
2	8	183	64	0	0	23.3	0
3	1	89	66	23	94	28.1	0
4	0	137	40	35	168	43.1	2

Next steps: Generate code with df New interactive sheet

```
df.shape
```

```
(768, 9)
```



```
df.head()
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunct
0	6	148	72	35	0	33.6	0
1	1	85	66	29	0	26.6	0
2	8	183	64	0	0	23.3	0
3	1	89	66	23	94	28.1	0
4	0	137	40	35	168	43.1	2

Next steps: Generate code with df New interactive sheet

```
df.shape
```

```
(768, 9)
```



```
df.columns
```

```
Index(['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insulin',
       'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome'],
      dtype='object')
```

```
df.dtypes
```

	0
Pregnancies	int64
Glucose	int64
BloodPressure	int64
SkinThickness	int64
Insulin	int64
BMI	float64
DiabetesPedigreeFunction	float64
Age	int64
Outcome	int64

dtype: object

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype  
---  ------                    --------------  -----  
 0   Pregnancies               768 non-null    int64  
 1   Glucose                   768 non-null    int64  
 2   BloodPressure             768 non-null    int64  
 3   SkinThickness             768 non-null    int64  
 4   Insulin                   768 non-null    int64  
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64  
 8   Outcome                   768 non-null    int64  
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
```

+ Code   + Text

```
df.isnull().sum()
```

	0
Pregnancies	0
Glucose	0
BloodPressure	0
SkinThickness	0
Insulin	0
BMI	0
DiabetesPedigreeFunction	0
Age	0
Outcome	0

dtype: int64

```
df.describe()
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	Diabet
count	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	
mean	3.845052	120.894531	69.105469	20.536458	79.799479	31.992578	
std	3.369578	31.972618	19.355807	15.952218	115.244002	7.884160	
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	
25%	1.000000	99.000000	62.000000	0.000000	0.000000	27.300000	
50%	3.000000	117.000000	72.000000	23.000000	30.500000	32.000000	
75%	6.000000	140.250000	80.000000	32.000000	127.250000	36.600000	
max	17.000000	199.000000	122.000000	99.000000	846.000000	67.100000	

```
df["Glucose"].mean()
```

```
np.float64(120.89453125)
```

```
df["Glucose"].median()
```

```
117.0
```

```
df["Glucose"].mode()
```

	Glucose
0	99
1	100

dtype: int64

```
df["Glucose"].max()
```

```
199
```

```
df["Glucose"].var()
```

```
1022.2483142519557
```

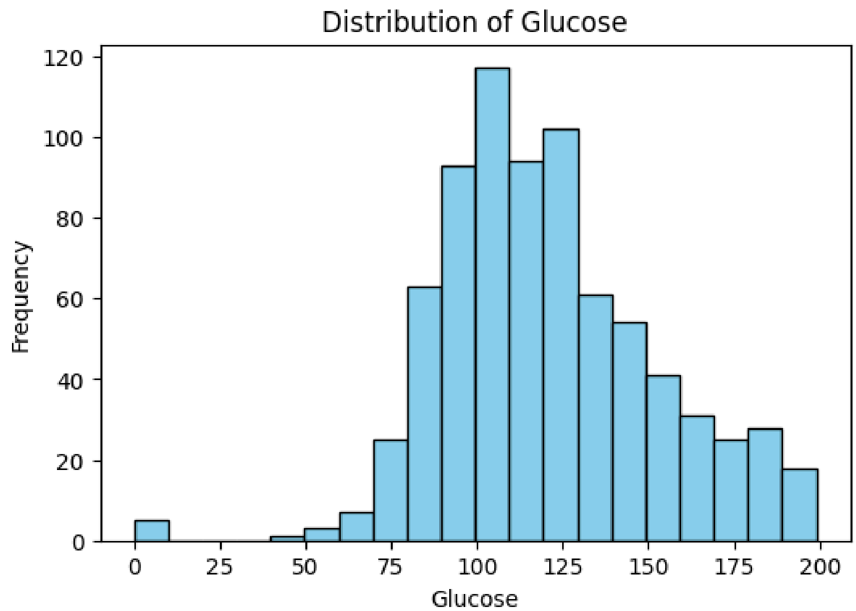
```
df["Glucose"].std()
```

```
31.97261819513622
```

```
columns = ["Glucose", "BMI", "Age"]

for col in columns:
    print("\nColumn:", col)
    print("Mean:", df[col].mean())
    print("Median:", df[col].median())
    print("Mode:", df[col].mode()[0])
    print("Minimum:", df[col].min())
    print("Maximum:", df[col].max())
    print("Variance:", df[col].var())
    print("Standard Deviation:", df[col].std())
```

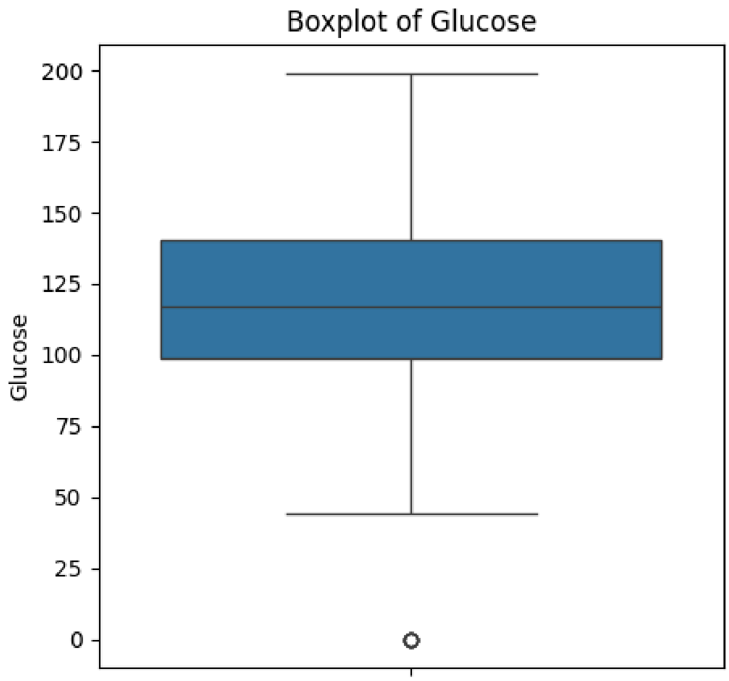
```
Column: Glucose
Mean: 120.89453125
Median: 117.0
Mode: 99
Minimum: 0
Maximum: 199
Variance: 1022.2483142519557
Standard Deviation: 31.97261819513622

Column: BMI
Mean: 31.992578124999998
Median: 32.0
Mode: 32.0
Minimum: 0.0
Maximum: 67.1
Variance: 62.15998395738257
Standard Deviation: 7.8841603203754405

Column: Age
Mean: 33.240885416666664
Median: 29.0
Mode: 22
Minimum: 21
Maximum: 81
Variance: 138.30304589037365
Standard Deviation: 11.76023154067868
```

```python
import matplotlib.pyplot as plt

plt.figure(figsize=(6,4))
plt.hist(df["Glucose"], bins=20, color="skyblue", edgecolor="black")
plt.title("Distribution of Glucose")
plt.xlabel("Glucose")
plt.ylabel("Frequency")
plt.show()
```

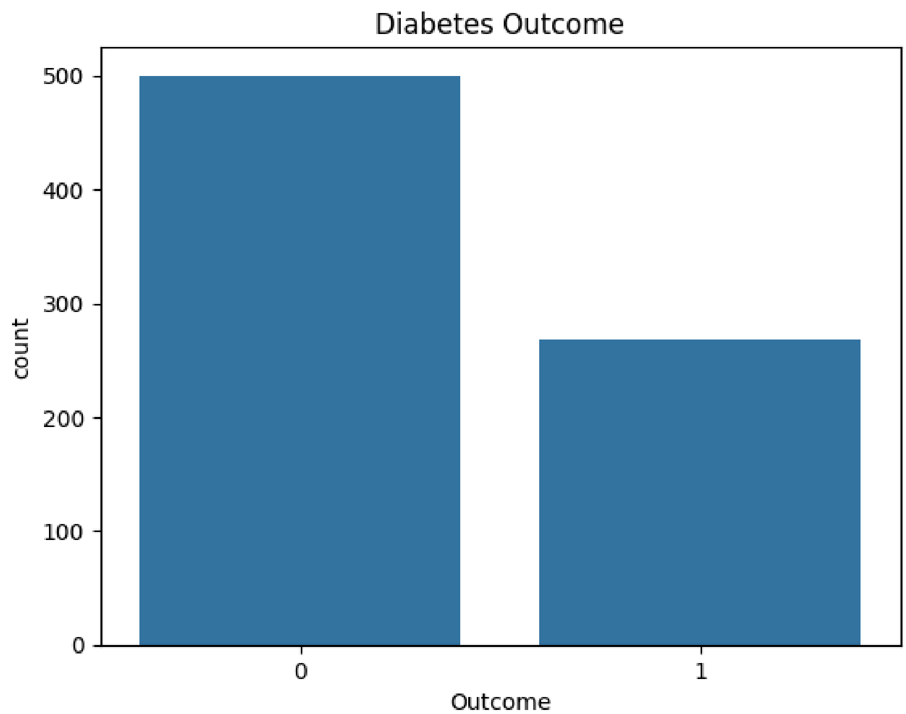



```python
import seaborn as sns

plt.figure(figsize=(5,5))
sns.boxplot(y=df["Glucose"])
plt.title("Boxplot of Glucose")
plt.show()
```

...

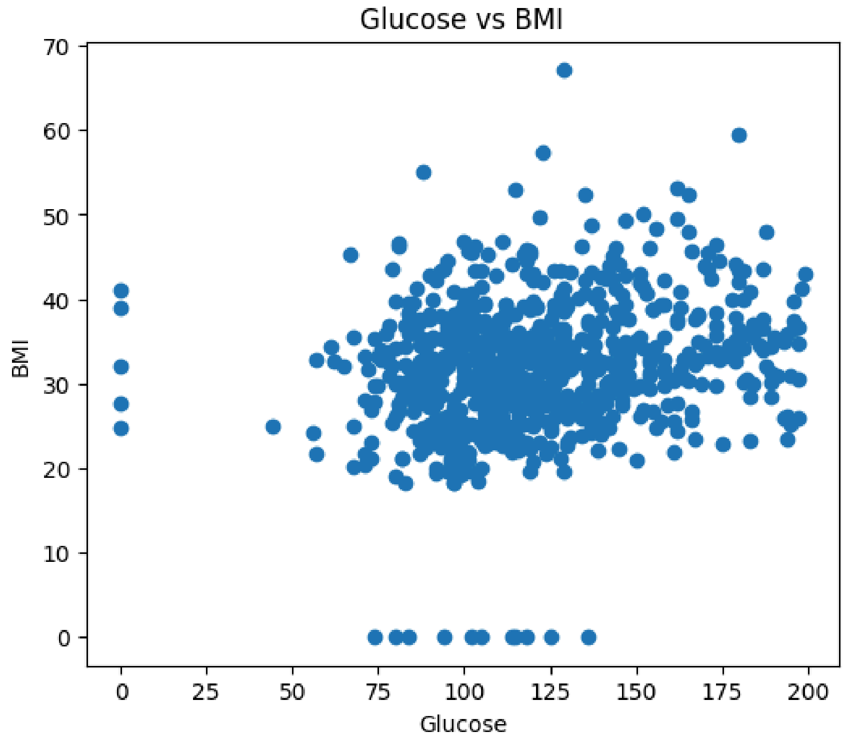



```python
sns.countplot(x="Outcome", data=df)
plt.title("Diabetes Outcome")
plt.show()
```



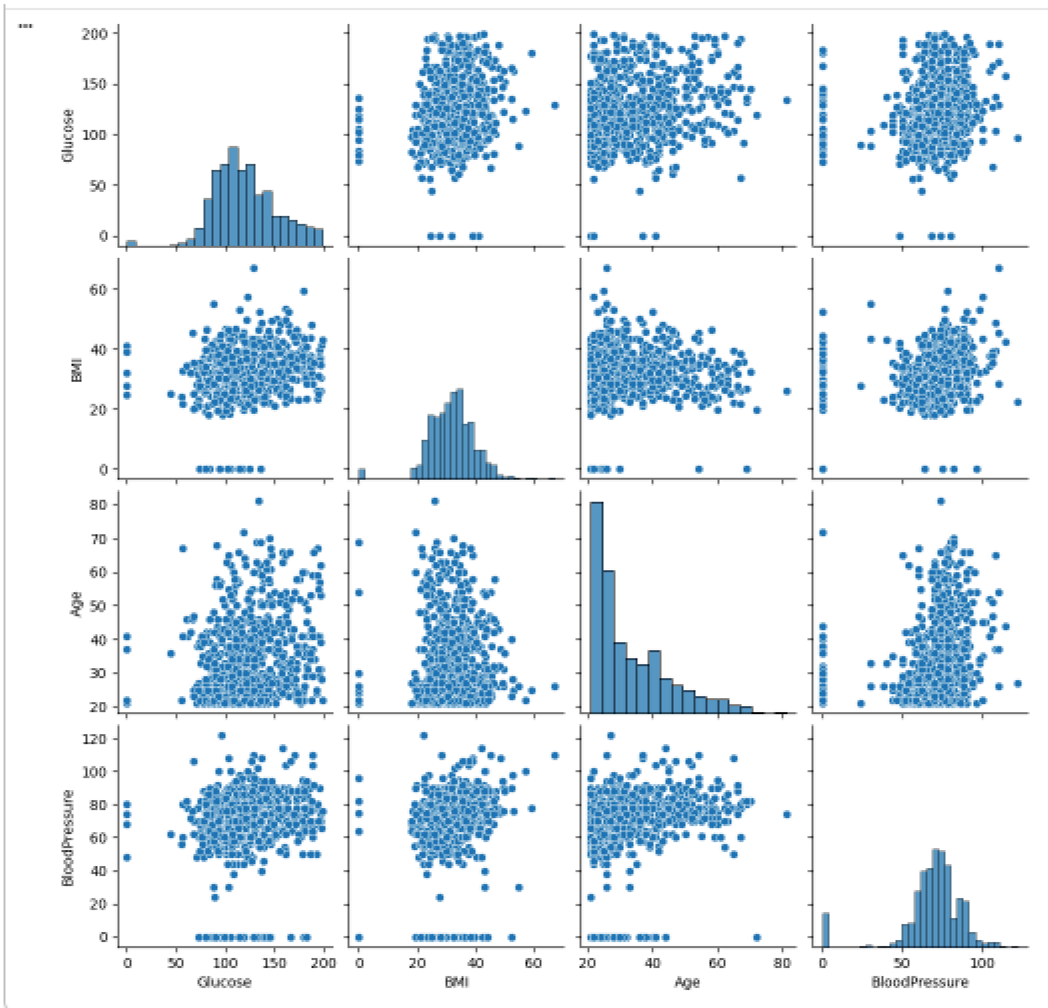

```
plt.figure(figsize=(6,5))
plt.scatter(df["Glucose"], df["BMI"])
plt.xlabel("Glucose")
plt.ylabel("BMI")
plt.title("Glucose vs BMI")
plt.show()
```

...




```
plt.figure(figsize=(6,5))
plt.scatter(df["Age"], df["Glucose"])
plt.xlabel("Age")
plt.ylabel("Glucose")
plt.title("Age vs Glucose")
plt.show()
```


```
sns.pairplot(df[["Glucose", "BMI", "Age", "BloodPressure"]])
plt.show()
```



Conclusion

Descriptive statistical measures and visualization techniques were applied to the Pima Indians Diabetes Dataset to understand the distribution, variability, and relationships among the variables.

Questions to be Answered by Students

1. Which numerical variable has the highest standard deviation?

ANS: Insulin has the highest standard deviation (approximately 115.24), indicating that insulin values are highly dispersed and vary significantly across patients.

2. Which variable shows the greatest difference between its mean and median?

ANS: Insulin shows the greatest difference between its mean and median.

Mean ≈ 79.8 , Median ≈ 30.5

This large difference indicates a positively (right) skewed distribution caused by many high insulin values and outliers.

3. What relationship can be observed between Glucose and BMI from the scatter plot?

ANS: The scatter plot shows a weak to moderate positive relationship between Glucose and BMI. In general, individuals with a higher BMI tend to have higher glucose levels, although the data points are fairly scattered, indicating the relationship is not very strong.

4. State two important insights obtained from the analysis.

ANS: i) Insulin values exhibit high variability and are heavily right-skewed, making them a candidate for preprocessing before model development.

ii) Higher Glucose and BMI values are more commonly observed among diabetic patients (Outcome = 1), suggesting these variables are strong predictors of diabetes.

5. Which variable(s), if any, contain potential outliers? How were these identified?

ANS: The following variables contain potential outliers:

- Insulin (most significant)
- BMI
- SkinThickness

- DiabetesPedigreeFunction
- BloodPressure (a few)

These outliers were identified using box plots, where observations lying outside 1.5 × IQR (Interquartile Range) beyond the first and third quartiles were considered outliers.

6. State **two important insights** obtained from your exploratory statistical analysis of the dataset.

   ANS: i) Several medical variables (Glucose, BloodPressure, SkinThickness, Insulin, and BMI) contain invalid zero values, indicating missing or incorrect measurements that require data cleaning before model training.

   ii) Glucose is the strongest indicator of diabetes, while BMI, Age, and Pregnancies also show positive associations with the Outcome variable, making them important features for diabetes prediction.